

Solving by Elimination Notes

Example 1: Solve the system of linear equations by elimination

$$\begin{array}{r} \begin{cases} x - 2y = 1 \\ 3x + y = 31 \end{cases} \quad \begin{array}{l} (x - 2y = 1) \cdot (-3) \\ 3x + y = 31 \\ \hline -3x + 6y = -3 \\ \hline 7y = 28 \\ \frac{7y}{7} = \frac{28}{7} \\ y = 4 \end{array} \end{array}$$

$$\begin{array}{l} x - 2(4) = 1 \\ x - 8 = 1 \\ +8 \quad +8 \\ \hline x = 9 \end{array}$$

$(9, 4)$

3 × 3 Elimination

Steps to Solve 3 × 3 Systems of Linear Equations by Elimination (Combination)

Process

1. Select a variable to eliminate.
Tip: Select a variable that can be eliminated with the fewest modifications to the original equations.
2. Combine any two equations - which may be multiplied by a constant - to eliminate the selected variable.
3. Combine a second set of equations - which may be multiplied by a constant - to eliminate the same variable.
4. Solve the new 2 × 2 system for both of its variables.
5. Substitute the values found in step 4 into any of the original equations to solve for the remaining variable.
6. Write the solution as an ordered triple, if appropriate.

Example 2: Solve the system of linear equations by elimination

$$\begin{array}{l} (3a - b + c = 5) \cdot (-2) \\ a + b + 2c = 15 \\ \hline -6a + 2b - 2c = -10 \\ \hline -5a + 3b = 5 \end{array}$$

$$\begin{cases} 6a - 2b + 2c = 10 \\ a + b + 2c = 15 \\ 3a - b + c = 5 \end{cases}$$

$$\begin{array}{l} (-6a - 2b + 2c = 10) \cdot (-1) \\ 6a - 2b + 2c = 10 \\ a + b + 2c = 15 \\ \hline -6a + 2b - 2c = -10 \\ \hline -5a + 3b = 5 \end{array}$$

$$\begin{array}{l} (-5a + 3b = 5) \cdot (-1) \\ -5a + 3b = 5 \\ 5a - 3b = -5 \\ \hline 0 = 0 \end{array}$$

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Example 3: Solve the system of linear equations by elimination

$$\begin{cases} x - y + 2z = 15 \\ -5y + z = 10 \\ 2x - 2y + 4z = 10 \end{cases}$$

$$\begin{array}{r} (x - y + 2z = 15) \cdot 2 \\ 2x - 2y + 4z = 10 \\ -2x + 2y - 4z = -30 \\ \hline 0 = -20 \end{array}$$

No Solution

Example 4: Solve the system of linear equations by elimination

$$\begin{array}{r} x + y + z = 0 \\ x - y + z = 8 \\ \hline \end{array}$$

$$2x + 2z = 8$$

$$\frac{2(17)}{2} + \frac{2z}{2} = \frac{8}{2}$$

$$\begin{array}{r} 17 + z = 4 \\ -17 \quad -17 \end{array}$$

$$z = -13$$

$$17 - y + (-13) = 8$$

$$4 - y = 8$$

$$-y = 4$$

$$y = -4$$

$$\begin{cases} x + y + z = 0 \\ 2x - y + z = 25 \\ x - y + z = 8 \end{cases}$$

$$\begin{array}{r} x + y + z = 0 \\ 2x - y + z = 25 \\ \hline \end{array}$$

$$(3x + 2z = 25) \cdot 1$$

$$\begin{array}{r} 3x + 2z = 25 \\ 2x + 2z = 8 \\ -3x - 2z = -25 \\ \hline \end{array}$$

$$-x = -17$$

$$x = 17$$

(17, -4, -13)